19/6/16

communication

Transmitting information from Transmitter to receiver

→ Analog communication.

Transforing of analog information from the Transmitter to the receiver.

→ Digital communication.

Transforing of digital information from the Transmitter to the receiver.

→ Discrete

→ digital.

→ short distance communication.

→ No need modulation

→ message is same as it is

→ Long distance communication

→ needs modulation.

→ Low freq message signal is modulated with the high freq carrier signal.

→ modulation.

Any one of the properties of carrier signal is change in accordance with message signal.

Analog modulation techniques:

→ Amplitude modulation, Frequency, phase

→ Both message and carrier is an analog signal.

AM:

The amplitude of the carrier signal is change in accordance with message signal.

Digital modulation techniques: [shift keying techniques]

→ ASK - amplitude shift keying
→ FSK - Frequency " "
→ PSK. - phase " "

→ message is digital and carrier in analog in nature.

ASK:

The amplitude of the carrier signal is change in accordance with digital msg signal.

nowadays all IC's are in digital IC's that why we use mostly use Digital communication.

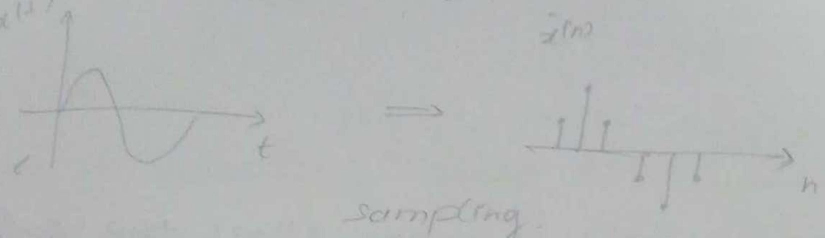
Importance of Digital comm.

→ most of the electronic gadgets and the computers use digital IC's for processing.

→ All practical signals are analog in nature and it must be converted to digital form for processing. Hence digital communication is use extensively nowadays.

Sampling:

→ It is the process of converting the analog signal into discrete signal.



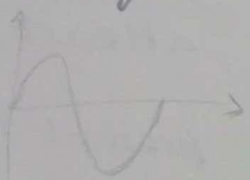
need for sampling:

→ All real time signals are analog in nature

→ Digital system cannot process analog signals hence analog signal is converted to discrete signal through sampling.

Sampling theorem:

$f(x) = a_0 + a_n \sum_{n=-\infty} \sin nx + b_n \sum \cos nx$



f_m → max freq of message signal

$\boxed{f_s \geq 2f_m}$

why time domain conv in freq domain

time domain → frequency domain ?

Bcz the mathematical operation is very easy in freq domain than the time domain.

$x_1(t) * x_2(t) \xrightarrow{F.T} X_1(j\omega) \cdot X_2(j\omega)$

Note:

1. $x(t) \xleftrightarrow{F.T} X(j\Omega)$ $\boxed{\Omega = \omega}$

2. $x(t) e^{j\Omega_0 mt} \xleftrightarrow{F.T} X[j(\Omega - m\Omega_0)]$

   └ Frequency shifting property

3. $\Omega = 2\pi f$

4. $\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT) \xleftrightarrow{F.T} \frac{1}{T} \sum_{m=-\infty}^{\infty} e^{jm\Omega_0 t}$

impulse train

Sampling theorem.

$$\frac{\pi}{T_s} \geq \Omega_m \Rightarrow f_s \geq 2f_m$$

$$\pi f_s \geq 2\pi f_m$$

$$\boxed{f_s \geq 2f_m}$$

Def: The sampling theorem gives the criteria of spacing 'Ts' between successive samples.

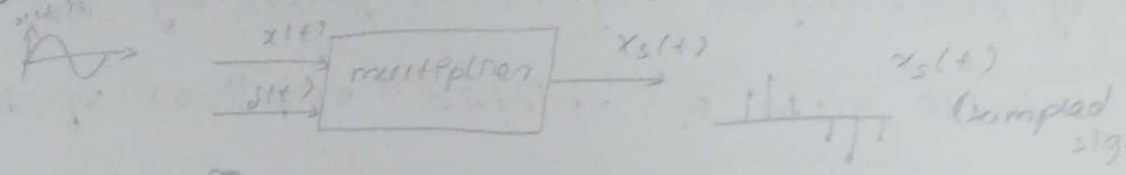
statement: A band limited signal $x(t)$ with $x(j\Omega) = 0$ for $\Omega > \Omega_m$. is uniquely determined from its sample if the sample frequency, $f_s \geq 2f_m$ that is, the Sampling frequency f_s must be atleast twice the maximum frequency present in the signal.

Proof of sampling theorem:

→ consider a continuous time signal $x(t)$ whose spectrum is bandlimited to Ω_m.

→ The sampled signal $x_s(t)$ can be obtain by multiplying $x(t)$ with an impulse train $\delta(t)$ with period T_s.

→ note the sampled signal take the period of impulse train



$-\infty \ldots ||| \ldots \infty \quad \delta_T(t)$.

* The bandlimited signal $x(t)$

$$x(t) \xleftrightarrow{FT} X(j\Omega)$$

* $x_s(t)$

$$x_s(t) = x(t) \cdot \delta_T(t)$$

$$= x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t-nT)$$

* $F[x_s(t)]$

$$F[x_s(t)] = F\left[x(t) \cdot \sum_{-\infty}^{\infty} \delta(t-mT)\right]$$

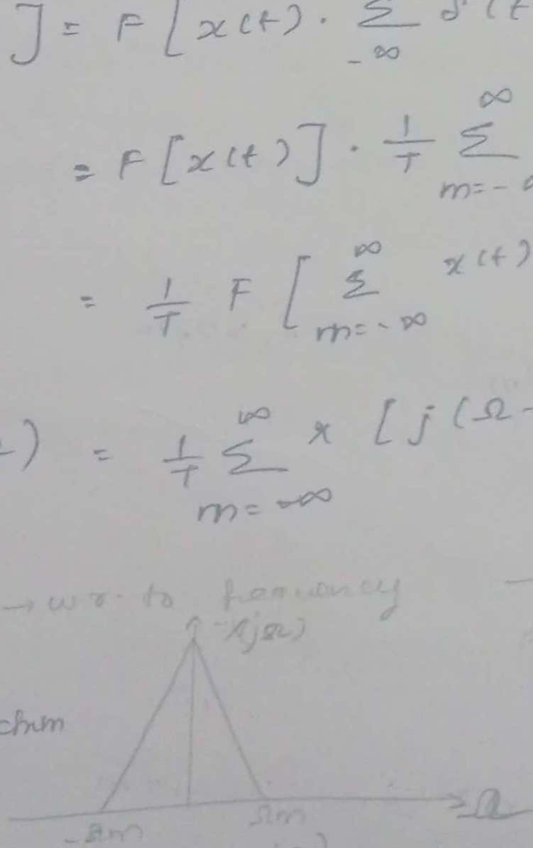
$$= F[x(t)] \cdot \frac{1}{T} \sum_{m=-\infty}^{\infty} e^{jm\Omega_0 t}$$

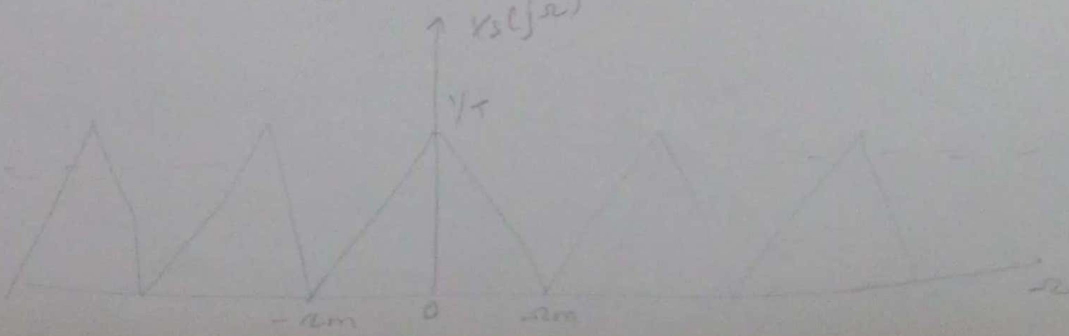
$$= \frac{1}{T} F\left[\sum_{m=-\infty}^{\infty} x(t) e^{jm\Omega_0 t}\right]$$

$$X_s(j\Omega) = \frac{1}{T} \sum_{m=-\infty}^{\infty} X[j(\Omega - m\Omega_0)] \longrightarrow ①$$

spectrum → w.r.to frequency

$x(t)$
message spectrum

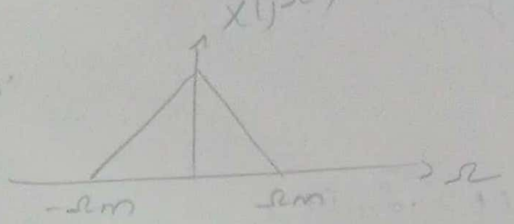




→ eqn ① reveals that the fourier transform of the sampled signal is given by an infinite sum of shifted replicas of fourier transform of the original message signal.

Aliasing

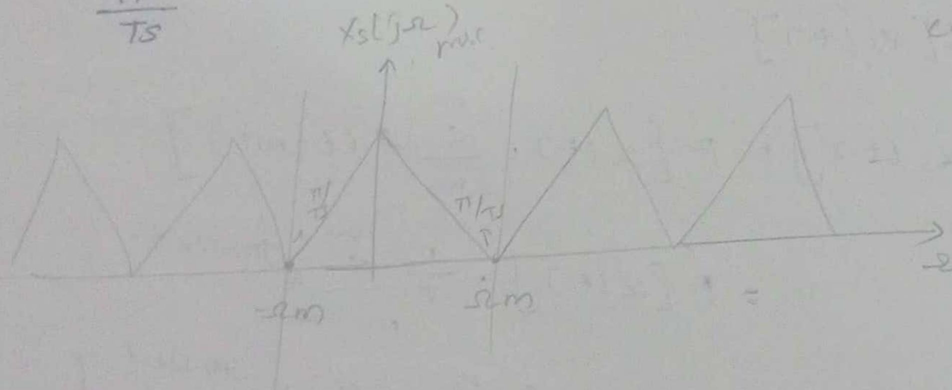
superposition of high freq & low freq components



msg sig

case(i)

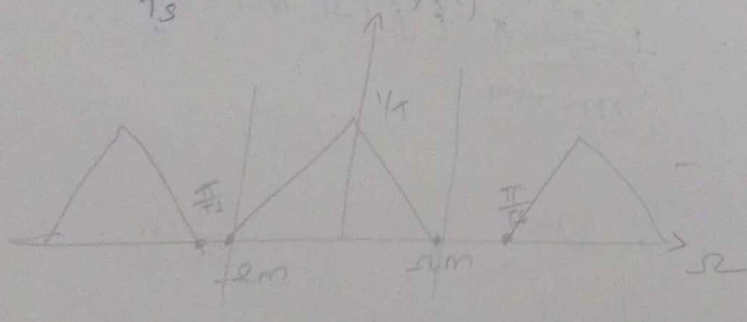
$\frac{\pi}{T_s} = \Omega m$ (or $f_s = 2f_m$)



can reconstruct msg sig

case(ii)

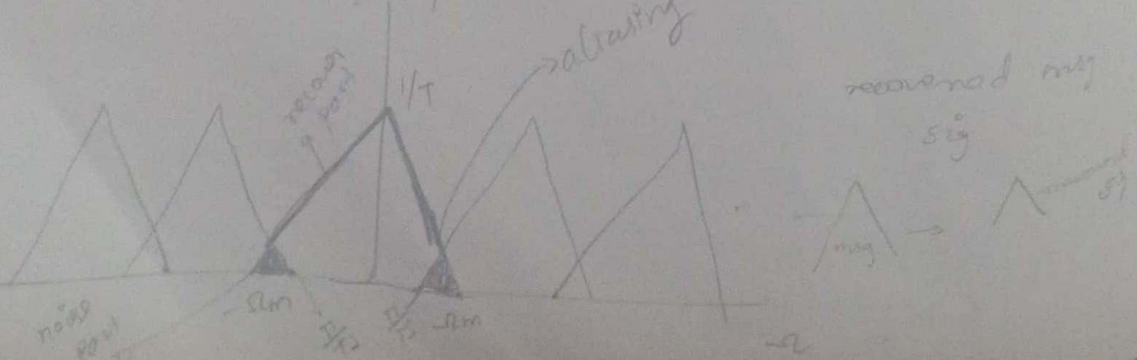
$\frac{\pi}{T_s} > \Omega m$ (or $f_s > 2f_m$)



can recover msg sig

case(iii)

$\frac{\pi}{T_s} < \Omega m$ (or $f_s < 2f_m$)



recovered msg sig

Conclusion

→ if $\frac{\pi}{T_s} \geq \Omega_m$ the replicas will not overlap and the message spectrum $X(j\Omega)$ can be recovered from $X_s(j\Omega)$ using a low pass filter with cutoff frequency $\Omega = \frac{\pi}{T_s}$.

→ if $\frac{\pi}{T_s} < \Omega_m$ the replicas will overlap and the message spectrum $X(j\Omega)$ cannot be recovered from $X_s(j\Omega)$

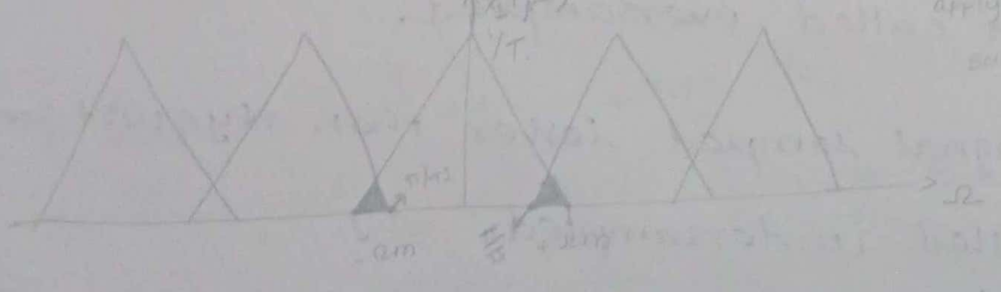
(avoid)
To overcome aliasing we should do
1. $f_s \geq 2f_m$
antialiasing filter (LPF)

analog i/p → LPF → Sampler

Aliasing:

→ The superimposition of high frequency component on low frequency component is known as frequency aliasing.

→ This happens when $\frac{\pi}{T_s}$ is less than Ω_m ($f_s < 2f_m$)

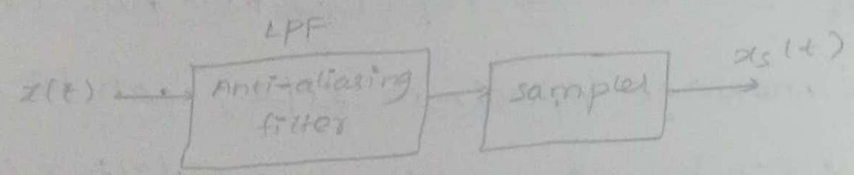


apply LPF to get bandlimited signal

→ Aliasing can be avoided by two methods

1. $f_s \geq 2f_m$. [Sampling frequency must be greater than or equal to twice of the maximum frequency present in the message signal]

2. By using an anti-aliasing filter.



Def of anti-aliasing filter.

A filter that is used to reject high frequency signal before it is sampled to reduce aliasing is called anti-aliasing filter.

Nyquist rate:

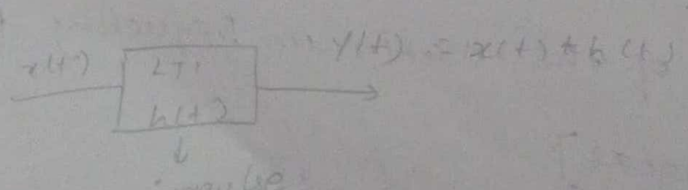
Nyquist rate → The minimum rate at which a signal is to be sampled and still we reconstructed is called Nyquist rate

$$\text{Nyquist rate} = 2f_m.$$

→ A signal sampled greater than Nyquist rate is called oversampled.

→ A signal sampled lesser than Nyquist rate is called undersampled.

NOTE:

1. 

2. convolution Integral

$$y(t) = x(t) * h(t)$$

$$= \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

3. convolution sum

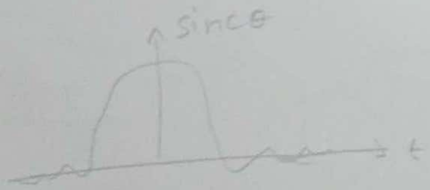
$$y(n) = x(n) * h(n)$$

$$y(n) = \sum_{m=-\infty}^{\infty} x(m)\, h(n-m).$$

* $e^{j\theta} = \cos\theta + j\sin\theta$

* $\sin\theta = \dfrac{e^{j\theta} - e^{-j\theta}}{2j}$

* $\text{sinc}\,\theta = \dfrac{\sin\theta}{\theta}$



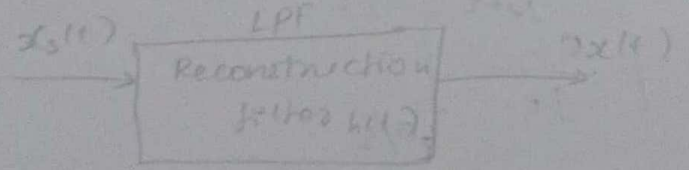
* IFT

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)\, e^{j\Omega t}\, d\omega$$

signal reconstruction.

All practical signals are causal signals.

A band limited signal [msg sig] $x(t)$ can be reconstructed from $x_s(t)$ by means of a reconstruction filter with impulse response $h(t)$



$$x(t) = x_s(t) * h(t)$$

$$= \sum x(nT)\, \delta(t-nT) * h(t)$$

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT) \int_{-\infty}^{\infty} \delta(\lambda - nT)\, h(t-\lambda)\, d\lambda.$$

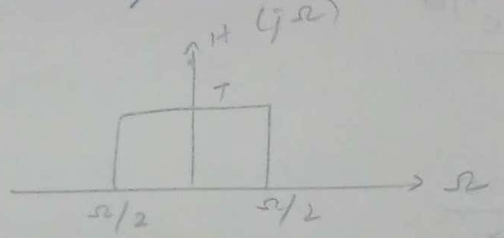
if $\lambda = nT$.

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT) \int_{-\infty}^{\infty} \delta(0)\, h(t-nT)\, d(nT)$$

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT)\, h(t-nT) \quad \longrightarrow (1)$$

Find $h(t-nT)$.

* spectrum of Recon-filter (LPF)



$$H(j\Omega) = \begin{cases} T & -\Omega/2 < \Omega < \Omega/2 \\ 0 & \text{elsewhere} \end{cases}$$

By IFT.

$$h(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} H(j\Omega)\, e^{j\Omega t}\, d\Omega$$

$$= \frac{1}{2\pi} \int_{-\Omega/2}^{\Omega/2} T \cdot e^{j\Omega t}\, d\Omega$$

$$= \frac{T}{2\pi} \left[\frac{e^{j\Omega t}}{jt} \right]_{-\Omega/2}^{\Omega/2}$$

$$= \frac{T}{2\pi jt} \left[e^{jt\Omega/2} - e^{-jt\Omega/2} \right]$$

$$h(t) = \frac{T}{\pi t} \sin\left(\frac{\Omega t}{2}\right) \qquad \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

WKT, $\Omega = \frac{2\pi}{T}$

$$h(t) = \frac{T}{\pi t} \sin\left(\frac{2\pi t}{T2}\right)$$

t → time domain

T → time interval

$$h(t) = \frac{T}{\pi t} \sin\left(\frac{\pi t}{T}\right)$$

$$= \frac{\sin(\pi t/T)}{\pi t/T}$$

$$h(t) = \text{sinc}\left(\frac{\pi t}{T}\right)$$

$$h(t-nT) = \text{sinc}\left[\frac{\pi(t-nT)}{T}\right] \rightarrow (2)$$

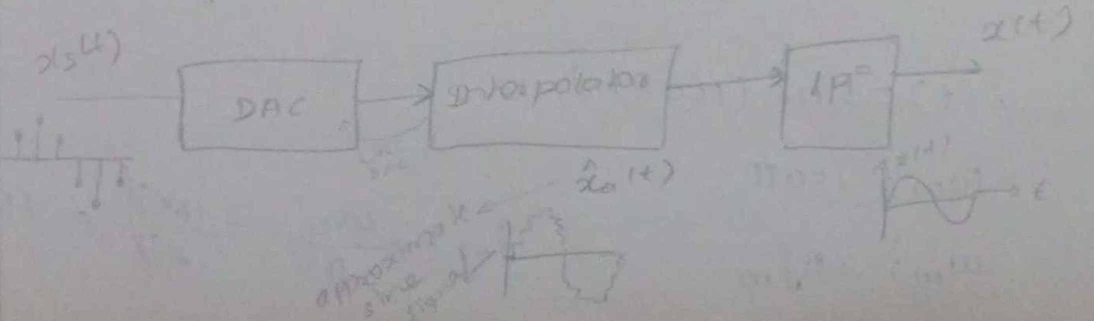
sub (2) in (1)

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT)\, \text{sinc}\left[\frac{\pi(t-nT)}{T}\right] \rightarrow (3)$$

<u>conclusion</u>

→ The response given by eqn (3) is not suitable for practical implementation. Because to reconstruct the msg signal at one time instant requires the knowledge of future values of $x(nT)$. So it represents a non-causal system.

→ For practical applications to reconstruct the message signal the below block diagram is used.



→ $x_s(t)$ is given to DAC, then gives to interpolator and it gives approx signal $\hat{x}_a(t)$ then it will move it on LPF

→ The o/p of LPF is the reconstructed msg signal.

<u>Problem:</u>

The signal having a spectrum ranging from DC to 10KHZ is to be sampled and converted to discrete form. what is the min. no. of samples per second (in HZ) that must be taken to ensure recovery.

<u>Soln</u> $f_s \geq 2f_m$.

freq of DC = 0

DC to 10KHZ

$f_m = 10KHZ$

$2f_m = 20KHZ$

$f_s \geq 20000$ HZ (or sample/sec)

A signal $x(t) = \text{sinc}(150\pi t)$, then sampled at the rate of,

a) 100HZ

b) 200HZ

c) 300HZ.

for each of these three cases can you recover the signal $x(t)$ from the sampled signal

<u>Soln</u> $x(t) = \text{sinc}(150\pi t)$

$\omega_m = 150\pi$

$\omega_m = 2\pi f_m$ $\qquad f_m = \frac{\omega_m}{2\pi} = \frac{150\pi}{2\pi} = 75$

$f_m = 75$, $2f_m = 150$.

a) $f_s = 100 Hz$

condition: $f_s \geq 2f_m$.

$100 \geq 150$

Not to be recovered because $f_s < 2f_m$

b) $f_s = 200 Hz$

$200 \geq 150$

This can be recovered because $f_s > 2f_m$

c) $f_s = 300 Hz$

$300 \geq 150$

This can be recovered because $f_s > 2f_m$

If $\sin c(150\pi t) + \sin(150\pi t) + \cos(100\pi t)$

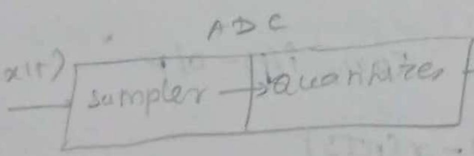
find max freq of $\underline{f_m}$. then proceed with conditions

---

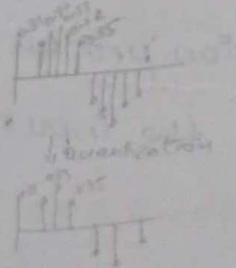
Quantizer:

change sampled values to some discrete values

ADC



advant	disadv
less memory space	accurate is less → quantization error



quantizer → ckt

quantization → process

Quantizer:

The circuit which performs the quantization process is called quantizer

Quantization:

The process of transforming the sampled amplitude values of an msg signal into a discrete amplitude value is refer to as Quantization.

Types of Quantizer

→ uniform Quantizer

→ non-uniform Quantizer

1. <u>uniform Quantizer (Linear Quantizer)</u>

→ Step size (δ) → constant

In uniform Quantization, the quantization step (difference between two Quantization levels) remains constant over the complete amplitude range

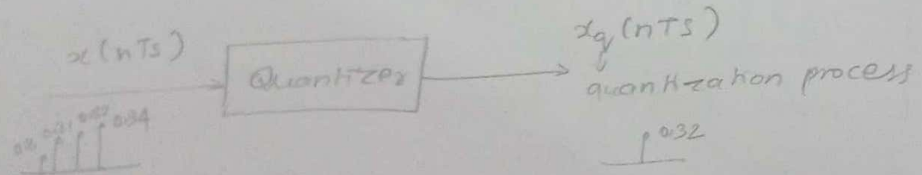
<u>Types</u>

→ mid-Tread Quantizer (Rounding)

→ mid riser " (Rounding)

→ Biased " (Truncation) Error

<u>Quantization Error</u>

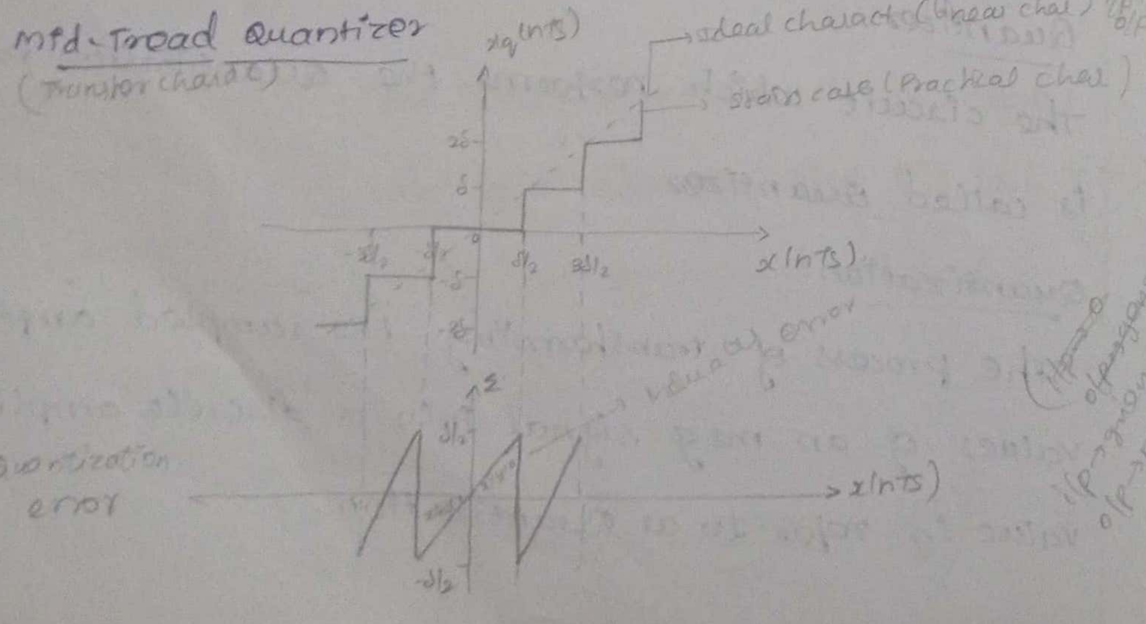


ε = o/p − i/p (or) i/p − o/p

$\varepsilon = x_q(nTs) - x(nTs)$

This error occurs because of Quantisation process. It is defined as the difference between the input and output

<u>Mid-Tread Quantizer</u>
(Transfer charact)



(i/p = o/p quantization)
i/p region
o/p is not quantized

⇒ The transfer characteristics of mid-Tread Quantizer is shown above. In the staircase like graph the origin lies in the middle of the Tread portion in mid-Tread type

⇒ when the i/p is between $-\delta/2$ to $\delta/2$, the output is zero; If $-\delta/2 < x(nT_s) < \delta/2$, then

$$x_q(nT_s) = 0$$

⇒ when i/p is between $\delta/2$ and $3\delta/2$ the output is δ

if $\delta/2 < x(nT_s) < 3\delta/2$ then

$$x_q(nT_s) = \delta$$

where δ is a step size

⇒ Similarly other levels are assigned. It is called mid-Tread quantizer because o/p is zero when $x(nT_s)$ is zero

⇒ Quantization error is given by,

$$\varepsilon = x_q(nT_s) - x(nT_s) \quad \text{[Gen. eqn]}$$

case (i) at origin

$x(nT_s) = 0$, $x_q(nT_s) = 0$

$\therefore \varepsilon = 0$

case (ii) $x(nT_s) \approx \delta/2$, $x_q(nT_s) = 0$,

$\varepsilon = \delta/2$.

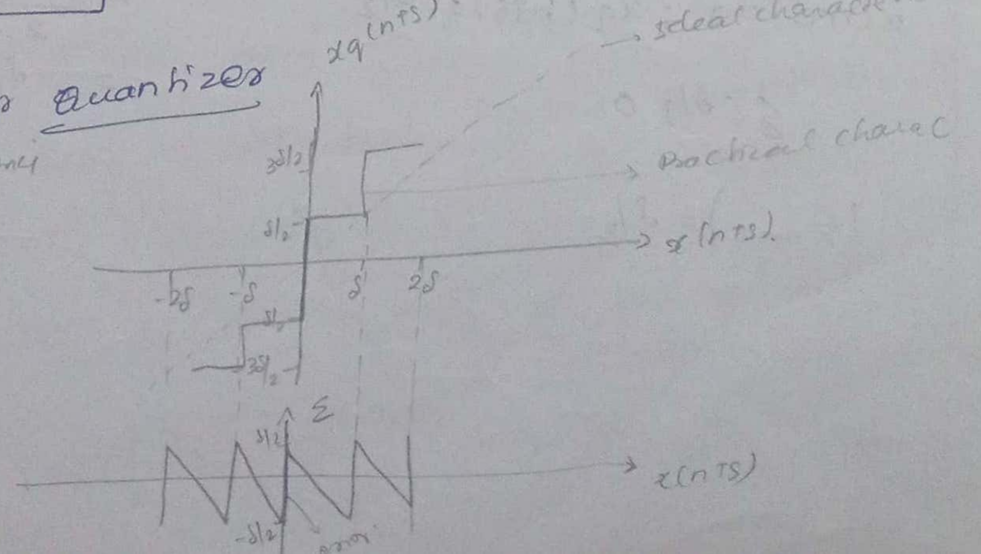
Max Quantization Error

i/p → a
o/p → generate

$$\boxed{|\varepsilon|_{max} = \delta/2}$$

Mid-riser Quantizer

Transfer characteristics



Quantization error

→ The transfer characteristics of mid-riser quantizer is shown above

→ When the input is zero then the o/p is becomes $\delta/2$ to $-\delta/2$.

If $x(nTs) = 0$ then,

$$\delta/2 < x_q(nTs) < \delta/2.$$

→ when the input is between δ and 2δ then the output is $3\delta/2$

$$x_q(nTs) = 3\delta/2$$

→ Similary the other levels are assigned. It is called mid-riser quantizer because of i/p is zero and output is $\delta/2$ & $-\delta/2$

⇒ Quantization error.

$$\varepsilon = x_q(nTs) - x(nTs)$$

case (i) at origin,

$x(nTs) \approx 0$; $x_q(nTs) = \delta/2$

$$\varepsilon = \delta/2$$

$x(nTs) = 0$; $x_q(nTs) = -\delta/2$

$$\varepsilon = -\delta/2$$

$$|\varepsilon_{max}| = \delta/2$$

case (ii)

$x(nTs) \approx \delta$, $x_q(nTs) = \delta/2$

$$\varepsilon = \delta/2 - \delta$$

$$= \frac{\delta - 2\delta}{2}$$

$$= -\delta/2$$

Biased Quantizer

→ Truncation process

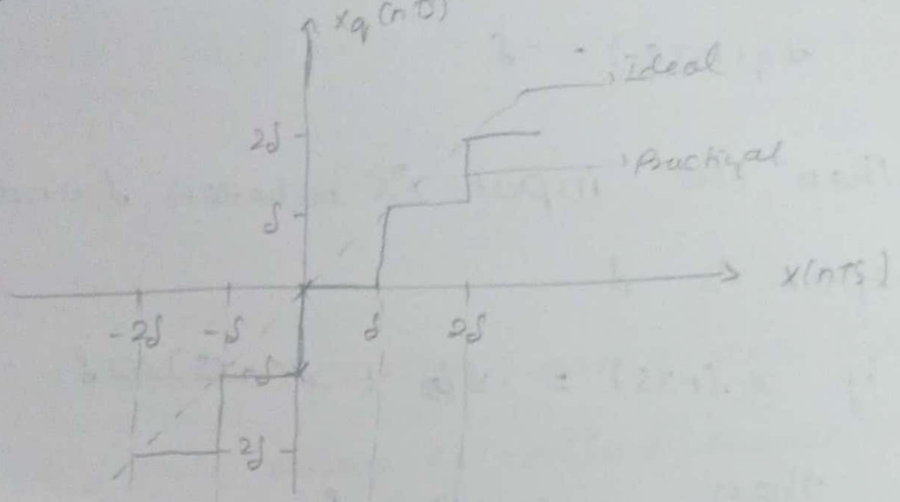
Rounding (3 decimal)

1.0035

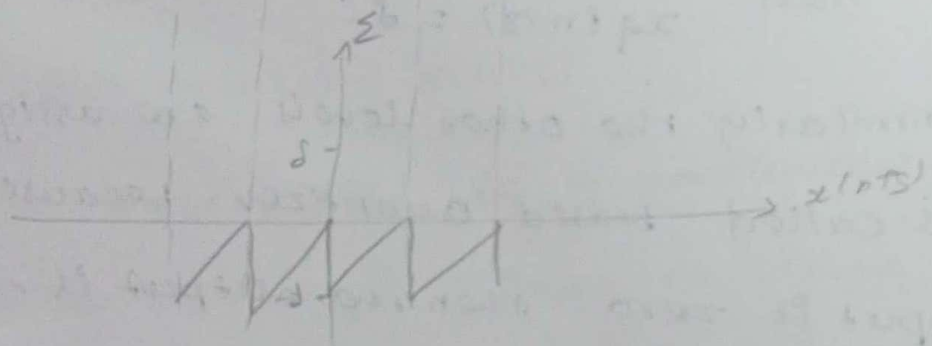
⇒ 1.004

Truncation → 1.003

Transfer characteristics



Quantization error



$$\varepsilon = x_q(nT_s) - x(nT_s).$$

At origin

(i) $x(nT_s) = 0$, $x_q(nT_s) = 0$

$\varepsilon = 0$

(ii) $x(nT_s) = 0$, $x_q(nT_s) = -\delta$

$\varepsilon = -\delta$

$|\varepsilon| = \delta$

$$|\varepsilon_{max}| = \delta$$

(ii) The mid-riser and mid-Tread quantizer are rounding quantizer but biased quantizer is the truncation quantizer.

(iii) The Transfer characteristics of the biased quantizer is shown above.

(iv) when the input is zero, then the output will be $-\delta$.

if $x(nTs) = 0$, then

$x_q(nTs) = -\delta$.

(v) When the input is between δ and 2δ the output is δ.

if $x(nTs) = \delta \le x(nTs) < 2\delta$

then $x_q(nTs) = \delta$.

similarly the other levels are assigned. It is called biased quantizer because if input is zero then the output is $-\delta$.

Conclusion

→ The quantization error of biased quantizer is more when compared to mid-Tread and mid riser quantizers.

Summary

1. Mid Tread quantizer.

quantization error, $|\varepsilon_{max}| = \delta/2$

2. Mid riser quantizer

quantization error, $|\varepsilon_{max}| = \delta/2$

3 Biased quantizer

quantization error, $|\varepsilon_{max}| = \delta$.

Normalization

not able to handle large values so that we can proceed with normalization means we choose the small values as per our convenient

Quantization noise and SNR in uniform Quantization (PCM)

SNR of uniform quant or SNR of PCM

Def: Because of Quanti

* step size: (δ).

→ Because of Quantization, errors are introduced in the signal. This error is called quantization error, $\varepsilon = x_q(nT_s) - x(nT_s)$

→ Let the input $x(nT_s)$, be of continuous amplitude in the range $-X_{max}$ to $+X_{max}$ then the total amplitude is $X_{max} - (-X_{max}) = 2X_{max}$

→ If there is 'q' quantization levels then the step size is,

q - quantization levels in quantizer

q=6

$$\boxed{\delta = \frac{2X_{max}}{q}}$$

→ If X_{max} is normalised to 1,

$$\boxed{\delta = \frac{2}{q}}$$

→ If 'v' is the no. of bits used in PCM then → represent the sampled signal

$q = 2^v$ and $\boxed{\delta = \frac{2}{2^v}}$

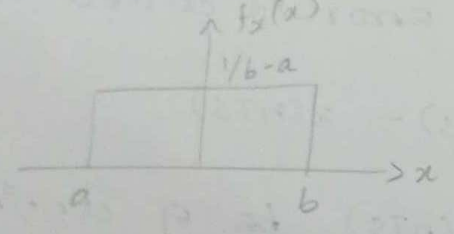
if v = 2 → 10

= 4 → 0010

* PDF of quantization error (ε)

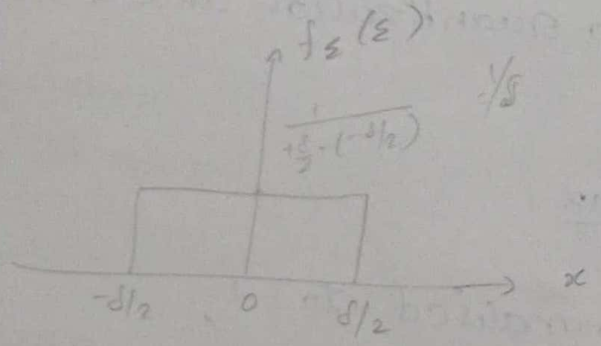
→ If the step size is sufficiently small, then it is reasonable to assume that the quantization error is uniformly distributed random variable.

→ Let us a mid-riser (tread) quantizer with the max quantization error is $\delta/2$.



→ Mean square value

$$E(x^2) = \int_{-\infty}^{\infty} x^2 f_x(x)\,dx.$$



$$f_\varepsilon(\varepsilon) = 1/\delta \quad ; \; -\delta/2 \text{ to } \delta/2$$

$$\qquad 0 \quad ; \text{ elsewhere}$$

* Noise power

$$SNR = \frac{\text{signal power (Normalized)}}{\text{Noise power (Normalized)}}$$

Noise power $= \frac{V^2}{R} \longrightarrow ①$

where $V^2 \rightarrow$ mean square of noise

$$V^2 = E(\varepsilon^2) = \int_{-\infty}^{\infty} \varepsilon^2 f_\varepsilon(\varepsilon) d\varepsilon$$

$$= \int_{-\delta/2}^{\delta/2} \varepsilon^2 \cdot \frac{1}{\delta} \, d\varepsilon$$

$$= \frac{1}{\delta} \left[\frac{\varepsilon^3}{3}\right]_{-\delta/2}^{\delta/2}$$

$$= \frac{1}{3\delta} \left[(\delta/2)^3 - (-\delta/2)^3\right]$$

$$= \frac{1}{3\delta}\left[\frac{\delta^3}{8} - \left(\frac{-\delta^3}{8}\right)\right]$$

$$= \frac{1}{3\delta}\left[\frac{\cancel{2}\delta^{\cancel{3}}}{\cancel{8}\,4}\right]$$

$$V^2 = E(\varepsilon^2) = \frac{\delta^2}{12}$$

apply in eqn ①

$$\text{Noise power} = \frac{\delta^2}{12R}$$

Assume R. is normalized to '1'

$$\boxed{\text{Noise power} = \frac{\delta^2}{12}}$$

* **SNR**

$$SNR = \frac{\text{sig. power}}{\text{Noise power}}$$

Let signal power be 'P'.

$$SNR = \frac{P}{\delta^2/12}$$

$$SNR = \frac{P}{\left(\frac{2X_{max}}{q}\right)^2 \times \frac{1}{12}} \quad , SNR = \frac{P}{(2/q)^2 \times \frac{1}{12}}$$

$$= \frac{P}{\left(\frac{2X_{max}}{2^v}\right)^2 \times \frac{1}{12}}$$

$$= \frac{P}{\frac{4X^2_{max}}{2^{2v}} \times \frac{1}{12_3}}$$

$$\boxed{SNR = \frac{3P2^{2v}}{x^2_{max}}}$$

assume $P=1$, $X_{max}=1$

$$\boxed{\therefore SNR = 3 \cdot 2^{2v}}$$

* In decibels

$$(SNR)_{dB} = 10\log(3 \cdot 2^{2v})$$

$$= 10\log(3) + 10\log(2^{2v})$$

$$(SNR)_{dB} = 10\log(3) + 2v \cdot 10\log(2)$$

$$= 4.771 + 2v(3.010)$$

$$(SNR)_{dB} = (4.8 + 6v)\,dB$$

where, v is the no. of bits represented by sampled signal, if P and X_{max} is normalised.

2/6/26 Limitation of uniform

* $SNR = \frac{\text{sig power}}{\text{noise power}}$

* SNR ↑

* In uniform quantization step size 'δ' is constant therefore noise power also be fixed

$SNR = \frac{2V}{1V}$ } not bearable

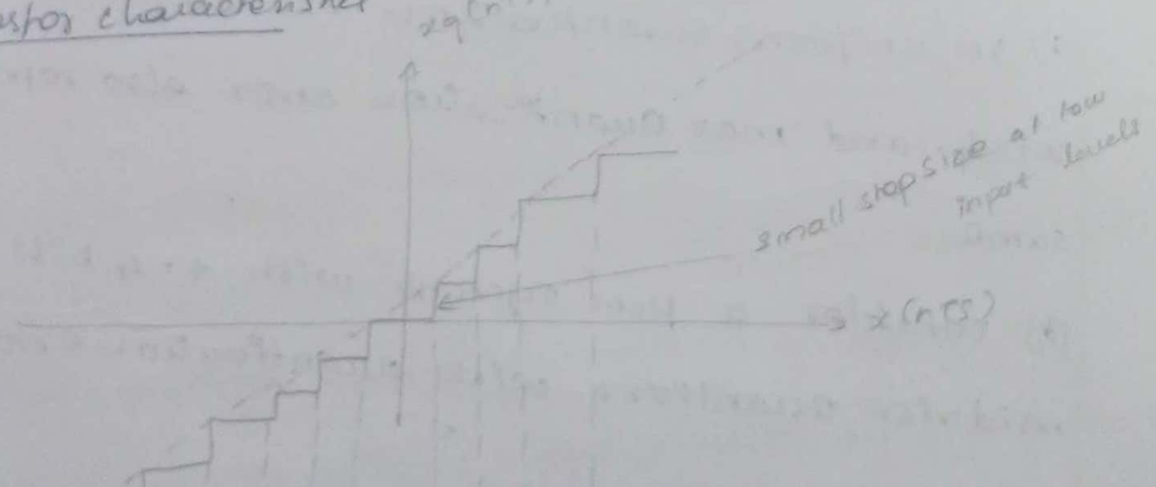
$SNR = \frac{16V}{1V}$ · bearable

→ Hence uniform quantization is suitable for large signals. not for small signals.

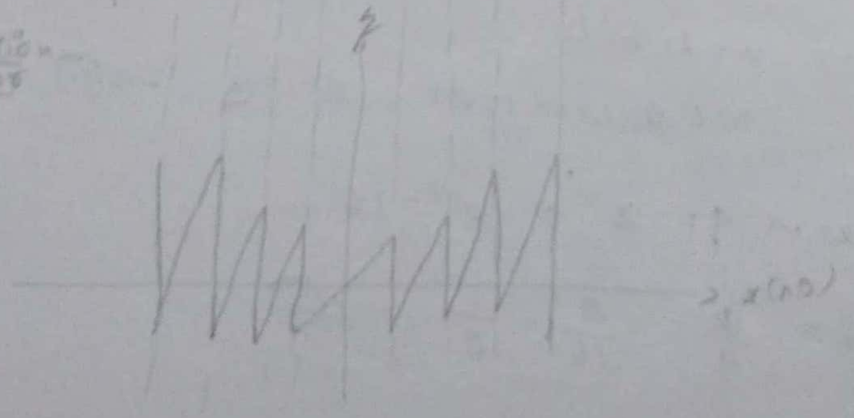
Non-uniform Quantization (include laws) (performance)

In non uniform quantization the step size is not fixed. It varies according to certain laws or as per input signal amplitude.

Transfer characteristic



Quantization error



→ In non-uniform quantization, the stepsize is small and low input signal levels. Hence quantization error is also small at these input levels. Therefore SNR is improve at low signal levels

→ As signal level increases the step size becomes higher and the noise increases hence SNR remains almost constant throughout the dynamic range of the quantizer.

Eg. $SNR = \frac{2mv}{0.25mv} = 8$ (small sig) both are same SNR

$SNR = \frac{16mv}{2mv} = 8$. (largest)

Need for non-uniform quantization.

i) In uniform quantization the step size remains const and max quantization error also remains same

(A) consider a PCM system with $v=4$ bits & midrise quantizer with quantization error $\delta/2$

PCM

$v = 4$ bits

MR quantizer ; $\epsilon = \delta/2$ → 1

PCM ⇒ $q = 2^v = 2^4 = 16$

$\delta = \frac{2}{q} = \frac{2}{16} = \frac{1}{8}$ → 2 sub ② in ①

$$\varepsilon = \frac{1}{16}.$$

$|\varepsilon|_{max} = \frac{1}{16}$ of max i/p amplitude range

consider an i/p with max amplitude of 16 volts.

$|\varepsilon|_{max} = \frac{1}{16} \times 16 = 1\,V$ = constant.

if the full range voltage is 16V, then quantization error is 1volts. which is tolerable. for low signal amplitudes like 2V, 3V etc the error of 1V is quite high. This problem arises bcz of uniform quantization.

→ stepsize vary, noise power vary, SNR constant

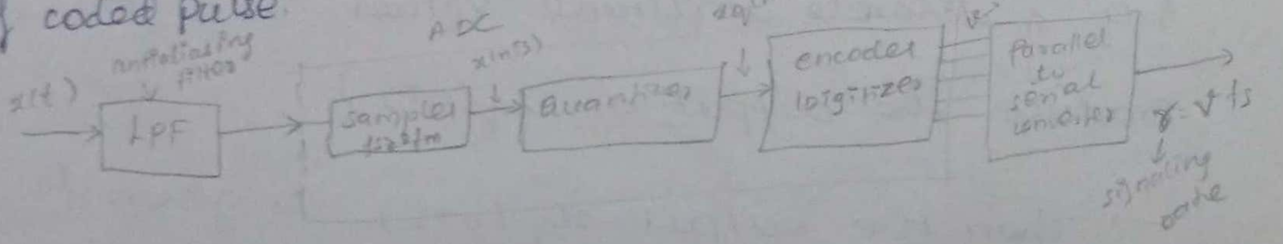
Conclusion: Therefore non-uniform quantization is used in such cases (low signal i/p levels)

(audio signals, speech signals) can use non-uniform quantization

↳ bcz amp is small

Pulse code modulation (PCM).

Def: PCM is a discrete time, discrete amplitude waveform coding process by means of which an analog signal is directly represented by a sequence of coded pulse.



PCM TX

Anti-aliasing filter:

→ The first block is used in this was the anti-aliasing filter which is nothing but the low pass filter.

→ It is a filter that is used to reject high frequency components

→ The output of this LPF is given to the first block of ADC namely the sampler block. The sampler block is helps to converts analog signal into discrete signals when sampling frequency (f_s) is greater than or equal to ~~max~~ twice of the maximum frequency of message signal [$f_s \geq 2f_m$].

→ Then the output of sampler is $x(nTs)$. and it is given as a input to the second block of ADC namely Quantizer. The process is done by Quantizer is called Quantization.

→ With the help of Quantizer (on used to rounds sampled amp to request predetermined discrete level) We can transforming the sampled amplitude values of an msg signal into a discrete amplitude values. Then it produce the output as $x_q(nTs)$

→ Then the output $x_q(nTs)$ is given to the third block of ADC as a input, namely encoder which is also called as digitizer. It is used to assign the equivalent binary values to the output of quantizer which is the predetermined discrete level.

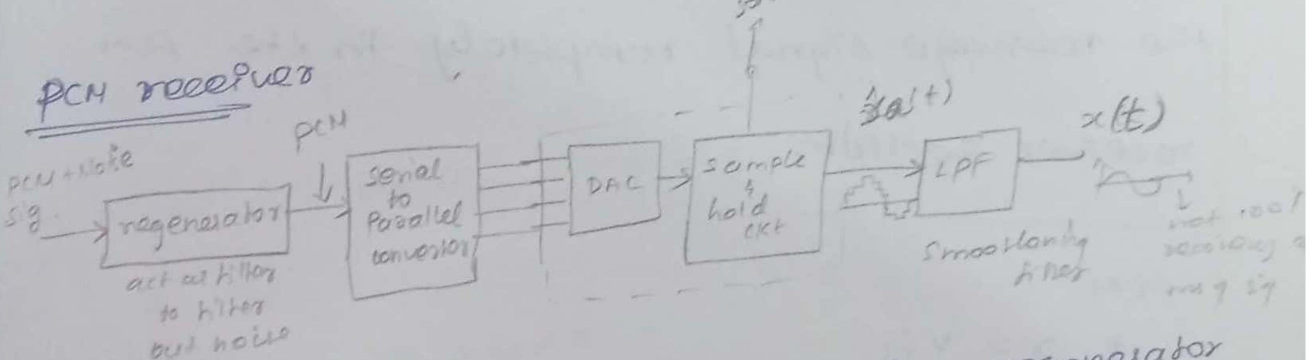
→ Last block of the PCM transmitter is Parallel to serial converter

→ It converts the parallel output of encoder to the serial output with signaling rate $r = v f_s$

$v \rightarrow$ no. of bits

$f_s \rightarrow$ sampling

PCM receiver



→ The first block of PCM receiver is regenerator which is act as a filter, it receives the PCM signal and noise is added with it up. To eliminate this noise, that's why we using regenerator block and give pure PCM signal

→ Then the PCM signal move to the second block which converts the PCM signal in the parallel form.

→ Then the o/p of serial to parallel converted is fed to the DAC, which is used to converts the digital signal into analog signal

→ Then move to the sample and hold circuit. it produces the approximate analog signal $\hat{x}_a(t)$,

→ Then $\hat{x}_a(t)$ is move to LPF which is act as smoothening filter. then produce the analog output $x(t)$.

→ Thus the message signal was reconstructed but in practically it won't be reconstructed the message signal completely in the PCM receiver section.

PCM:

* $r = v f_s$
* $BW \geq r/2$

$$BW \geq \frac{v f_s}{2}$$

$$BW \geq \frac{v \cancel{2} f_m}{\cancel{2}}$$

$$\boxed{BW \geq v f_m}$$

1. A Television signal with a BW of 4.2 MHZ is transmitted using Binary PCM. No. of Quantization levels is 512 calculate

(i) code word length (v)

(ii) final bit rate

(iii) Transmission BW.

(iv) output SNR.

soln Given:

$BW = 4.2 MHZ$

$q = 512$

(i) code word length

$q = 2^v$

$512 = 2^v$

$\log(512) = \log 2^v$

$2.7092 = v$

$\log 2.7092 = v \log 2$

$v = 8.999$

$\boxed{v = 9 \text{ bits}}$

(ii) Final bit rate (signaling rate)

$r = v f_s$

$BW = 4.2 MHZ$

$BW = f_H - f_L$ or $f_2 - f_1$

$BW \approx f_H$ (or) f_2

$\therefore f_H$ (or) $f_2 = 4.2 MHZ = f_m$.

$\therefore f_m = 4.2 MHZ$

$f_s = 2f_m = 2(4.2 MHZ)$

$f_s = 8.4 MHZ$

$r = 9(8.4 MHZ)$

$r = 75.6 \times 10^6$ HZ. (or)

75.6×10^6 samples/sec

$r = 75.6 \times 10^6$ bits/sec

(iii) Transmission BW.

$BW \geq r/2$

$BW \geq 31.8$ MHZ.

(iv) output SNR

$SNR = 3 \times 2^{2V}$

$= 3 \cdot 2^{2V}$

$= 3 \cdot 2^{2(9)}$

$= 3 \cdot 2^{18}$

$= 786432$

$(SNR)_{dB}$

$(SNR)_{dB} = (4.8 + 6V)_{dB}$

$= 4.8 + 54$

$(SNR)_{dB} = 58.8$

3/7/25

1. The bandwidth of a signal input to the PCM is restricted to 4KHZ. The input varies from -3.8V to +3.8V and has average power of 30mW The required SNR is 20dB. The modulator produces binary output Assume uniform quantization.

(i) calculate the no. of bits required per sample

(ii) output of 30 such PCM channel are time multiplexed. what is the min required transmission bandwidth for the multiplexed signal

soln Given:

BW = 4KHZ

$x_{max} = -3.8V$ to $+3.8V$

signal Power = 30 mW

$(SNR)_{dB} = 20$ dB

$10 \log (SNR)_{dB} = 20$ dB

(i) $\log(SNR) = 2. \rightarrow SNR = \text{antilog}(2) \rightarrow (10)^2$

$SNR = 100$

WKT,

$$SNR = \frac{3 \cdot P \cdot 2^{2V}}{x^2_{max}}$$

$$\frac{(100)\, x^2_{max}}{3 \cdot P} = 2^{2V}$$

$$\frac{100\,(3.8)^2}{3\,(30m)} = 2^{2V}$$

$2^{2V} = 2^2 \cdot 2^V$

$2^2 \cdot 2^V = 2^{(2+V)}$

$(2^2)^V = 2^{2V}$

$$16044.4 = 2^{2V}$$

$$\log(16044.4) = \log(2)^{2V}$$

$$4.205 = 2V \log 2$$

$$\frac{4.205}{0.3010 \times 2} = V$$

$V = 6.9843$ bits

$\boxed{V = 7 \text{ bits}}$

(ii) Transmission Band width of one pcm channel.

$BW = r/2$

$$BW = \frac{Vf_s}{2}$$

$f_m = 4KHZ$, $f_s = 2f_m = 8KHZ$

$$BW = \frac{V(2f_m)}{2}$$

$BW = V \cdot f_m$

$= 7\,(4KHZ)$

$BW = 28KHZ$

Transmission bw of 30 pcm channel.

$$1BW \times 30 = (28\,KHZ)\,30$$

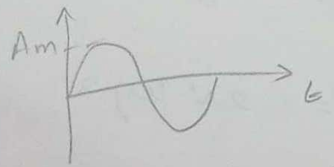
$$= 840\,KHZ$$

Prove that SNR value of sine signal

A PCM system uses a uniform quantizer followed by 'v' bit encoder. Show that signal to quantization noise ratio is approximately given by $(1.8 + 6v)_{dB}$ sine wave

$(4.8 + 6v)_{dB}$ → for general waveform

* consider a sinusoidal signal with amplitude 'Am'.



$$\text{Rms value} = \frac{Am}{\sqrt{2}}$$ → for sine signal

* ~~SNR~~ ~~find~~.

* $$\text{signal power } (P) = \frac{V^2 \text{signal}}{R}$$

assume R is normalized to 1

$$\text{signal power} = V^2 \text{signal}.$$

where V^2signal = mean square of signal

* $$SNR = \frac{3 \cdot P\, 2^{2V}}{x^2 max}$$

$$SNR = \frac{3 \cdot \left(\frac{Am}{\sqrt{2}}\right)^2 2^{2V}}{x^2 max}$$

WKT, $x\,max = Am$

$$SNR = \frac{3\left(\frac{Am^2}{2}\right) 2^{2V}}{Am^2}$$

$$SNR = 1.5\, 2^{2V}$$

$$\cancel{SNR = 10\log(1.5 \cdot 2^{2V})}$$

$$\cancel{= 10(2V)\log(1.5\times 2)}$$

$$\cancel{= 20V\log(3)}$$

$$\cancel{(SNR)_{dB} = \cancel{V}\, 0.477}$$

$$\cancel{= 9.542\, V.}$$

Take log on both sides.

$$SNR = 10\log(1.5 \cdot 2^{2V})$$

$$= 10\left[\log(1.5) + \log(2^{2V})\right]$$

$$= 10\log(1.5) + 2V\log(2)$$

$$= 1.76 + 6V$$

$$(SNR_{dB}) = (1.8 + 6V)\, dB$$

$\therefore$ Hence proved.

SNR for any signal,

$$(SNR)_{dB} = (4.8 + 6V)\, dB$$

SNR for sine signal,

$$(SNR)_{dB} = (1.8 + 6V)\, dB$$

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Assignment - 1.

Logarithmic companding of speech signal:

companding:

companding is the process of compressing the amplitude of a signal before quantization and transmission, and expanding it after reception to reduce quantization noise and improve the quality of low-level signals.

Logarithmic companding:

Logarithmic companding is a technique used in digital communication systems to improve the quality of speech signals. It compresses the dynamic range of the speech signal before transmission and expands it back to its original form at the receiver.

Need for companding:

speech signals have a wide range of amplitudes. If a uniform quantizer is used:

→ Low-amplitude signals suffer from high quantization noise

→ High-amplitude signals are represented accurately.

Types of Logarithmic companding:

1. μ-law (MU-Law) companding

2. A-law companding.

(1) μ-law companding:

μ-law companding is a logarithmic companding technique used to compress the dynamic range of a speech signal before quantization and expand it after reception. It improves the signal-to-Quantization Noise ratio (SQNR) and is mainly used in North America and Japan.

→ The standard compression parameter is $\mu = 255$

compression formula:

$$y = \mathrm{sgn}(x) \frac{\ln(1+\mu|x|)}{\ln(1+\mu)}$$

where,

x = Normalized input signal $(-1 \le x \le 1)$

y = compressed output signal

$\mu = 255$

$\ln$ = Natural logarithm

$\mathrm{sgn}(x)$ = sign of the input signal.

(ii) A-law companding

A-law companding is a logarithmic companding technique used to compress the dynamic range of a speech signal before quantization and expand it after reception. It reduces quantization noise and is mainly used in Europe and India.

compression formula:

$$y = F(x) = \frac{sgn(x)}{1+\ln(A)} \begin{cases} A|x| & , 0 \le |x| \le \frac{1}{A} \\ 1+\ln(A|x|) & , \frac{1}{A} \le |x| \le 1 \end{cases}$$

where,

x = Normalized input signal

y = compressed output signal

$A = 87.6$

$\ln$: Natural logarithm

$sgn(x)$ = sign of the input signal.

⇒ The standard compression parameter is $A = 87.6$

2. Time Division Multiplexing:

TDM is the process in which the number of signals are made to pass through a common channel at different time slots. A TDM system consists of the following parts:

N number of message signals are given to N LPFs, a commutator, pulse amplitude modulator, communication channel, pulse amplitude demodulator, Decommutator and Reconstruction low pass filter.

Working Principle:

Each input message signal is given to a LPF to remove the frequencies that are not essential to an adequate signal representation.

The LPF output are then applied to a commutator, which is usually implemented using electronic switching circuitry.

The function of commutator is two fold.

i) Since it is a rotating switch, it extracts one sample from each input per revolution i.e., it takes a narrow sample of each of the N input message at a rate f_s that is higher than $2w$ where, $w \rightarrow$ is the cutoff frequency of prealias filter

ii) To sequentially interleave these N-samples inside the sampling interval T_s. This is the main function of TDM.

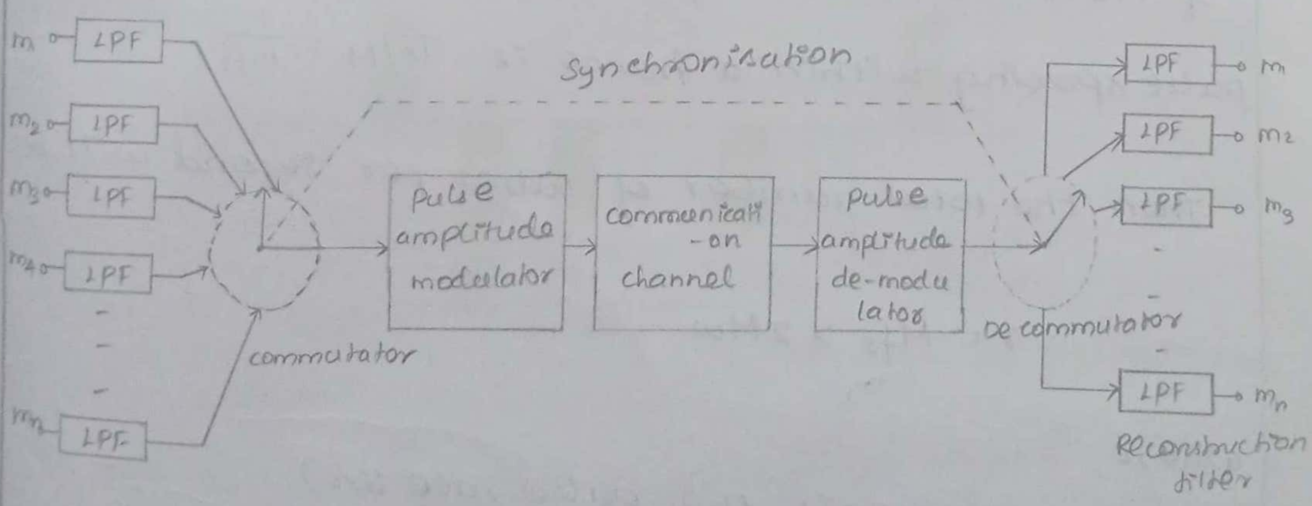
→ Next to this commutation process, the multiplexed signal is applied to pulse modulator. The purpose of pulse modulation is to transform the multiplexed signal into a form suitable for transmission over the communication channel.

→ A TDM system introduces a bandwidth expansion factor N, because the scheme squeeze N samples derived from N independent message sources into a time slot equal to one sampling interval

→ At the receiving of the system the received signa is applied to a pulse demodulator, which performs the reverse operation of the pulse modulator. The narrow samples produced at the pulse demodulator output are distributed to the approximate lowpass reconstruction filters by means of a demodulator

→ This decommutator operates in synchronism with commutator in the transmitter. This synchronisation is essential for satisfactory operation.

Block diagram:



→ TDM is highly sensitive to dispersion in the common channel i.e., to variations of amplitude with frequency (or) lack of proportionality of phase with frequency.

→ Accurate equalisation of both amplitude and Phase responses of the channel is necessary to ensure (or) perform a satisfactory operation of the system

→ TDM is immune to amplitude linearities in the channel as a source of cross-talk, because the different message signals are not simultaneously impressed on the channel.

→ If all the input have the same bandwidth w, the commutator should rotate at the rate $f_s \geq 2w$, so that successive samples from any one input are spaced by $T_s = 1/f_s \leq 1/2w$. The time interval T_s containing one sample from each input is called a frame.

If there are one M input channels, the pulse to pulse spacing within a frame is $T_s/M = \frac{1}{Mf_s}$.

Then the total number of pulses per second will be,

$$r = Mf_s \geq 2Mw$$

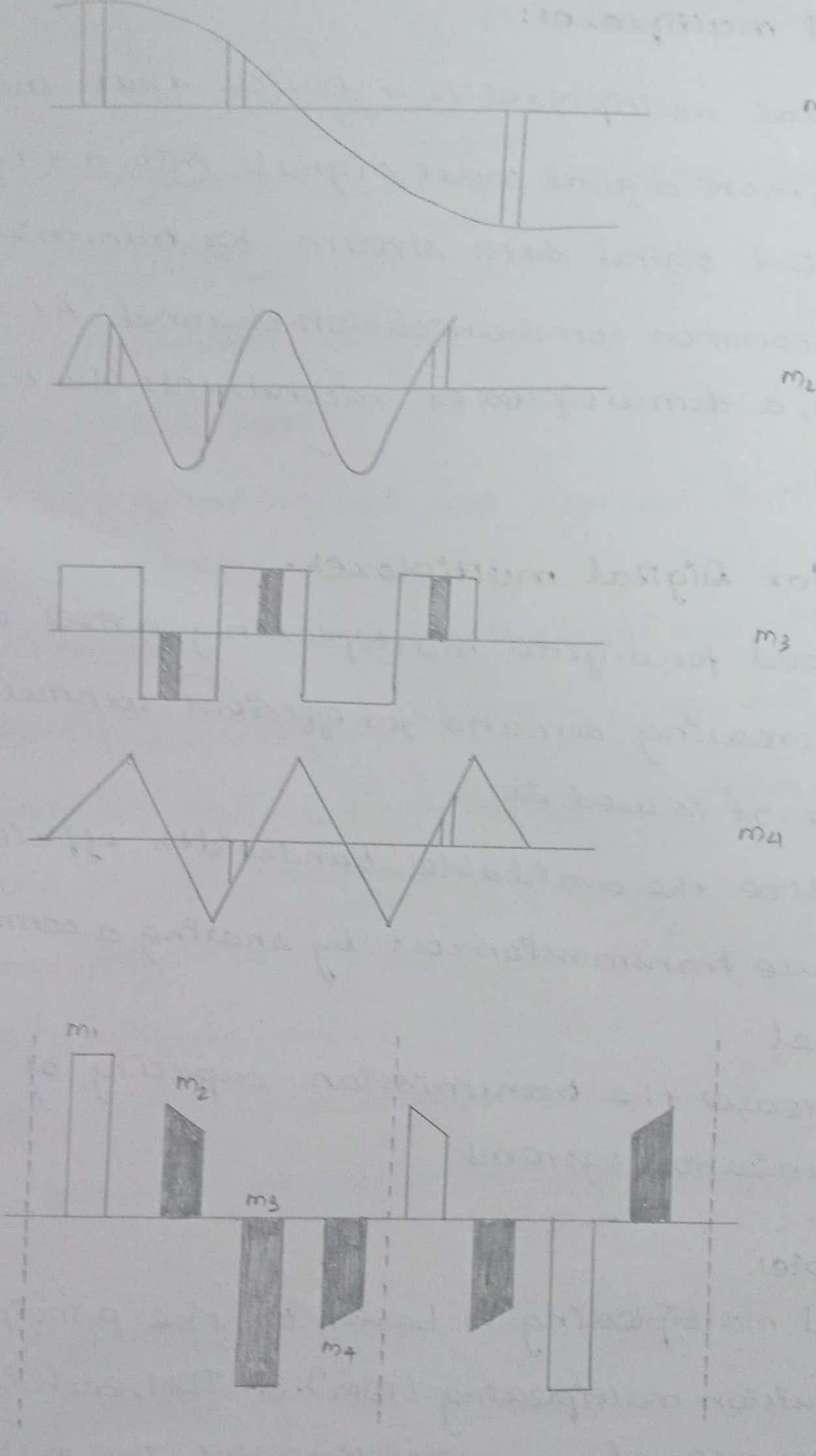
where,

$r \Rightarrow$ represents the pulse rate (or)

signaling rate of the TDM signal

Let us consider four messages as shown,

m_1, m_2, m_3, m_4



Waveforms showing multiplexed signal.

3- Digital multiplexer:

A digital multiplexer is a device that combines two or more digital input signals into a single high-speed digital data stream for transmission over a common communication channel. At the receiver, a demultiplexer reconstructs the original signals.

Need for Digital multiplexer.

The need for digital multiplexing arises due to the increasing demand for efficient communication systems. It is used to:

→ utilize the available bandwidth efficiently

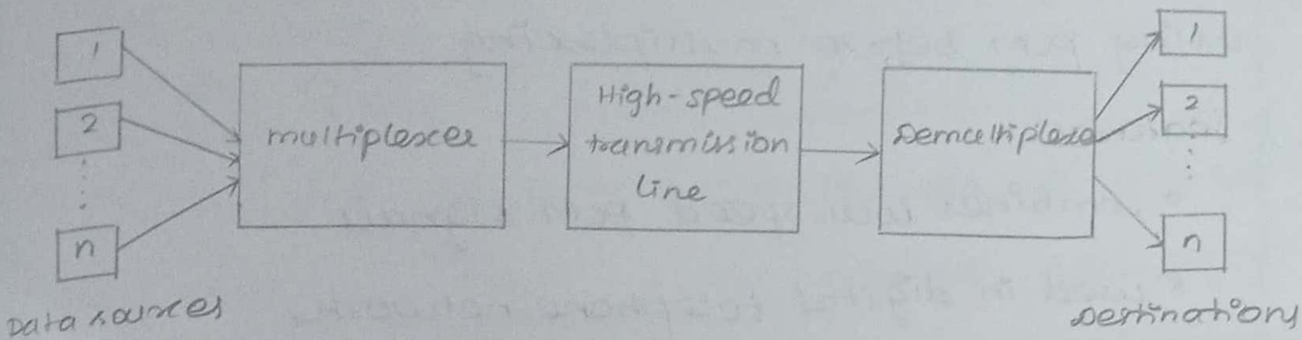
→ Reduce transmission cost by sharing a common channel

→ Increases the transmission capacity of communication systems

Principle:

Digital multiplexing is based on the principle of Time division multiplexing (TDM). In TDM, each input signal is assigned a specific time slot. The multiplexer sequentially selects one bit or one group of bits from each input signal and combines them into a single output stream. At the receiver, synchronization allows the demultiplexer to identify each time slot and recover the original signals.

Block diagram:



Working:

→ multiple digital signals are applied to the inputs of the multiplexer

→ The multiplexer selects each signal one after another according to the assigned time slots

→ The selected bits are combined to form a single high-speed digital bit stream

→ This multiplexed signal is transmitted through a common communication channel.

→ At the receiver, the demultiplexer uses synchronization information to seperate the incoming data

→ Each seperated signal is delivered to its corresponding output.

Types of Digital multipliers:

(1) Low Bit-Rate Digital multiplexer:

These multiplexers combine several low-speed digital signals into one higher-speed digital signal. They are commonly used in telephone systems where voice signals are converted into digital form using PCM before multiplexing.

features:

* combines low-speed PCM signals
* used in digital telephone networks
* supports bit rates up to a few megabits per second.

(ii) High-Bit-rate digital multiplexer.

High bit-rate multiplexers combine already multiplexed signals to obtain very high transmission rates. They are used in optical fiber communication, satellite communication, and long-distance communication systems.

features:

* combines high-speed digital streams
* used in backbone communication networks
* provides high transmission capacity.

Applications:

* Digital telephone systems
* optical fiber communication
* satellite communication
* Internet backbone networks
* Digital television broadcasting.